Zero-shot imputation of coastal in-situ chlorophyll-*a* gaps

Tongoy Balsa, Region de Coquimbo, Chile

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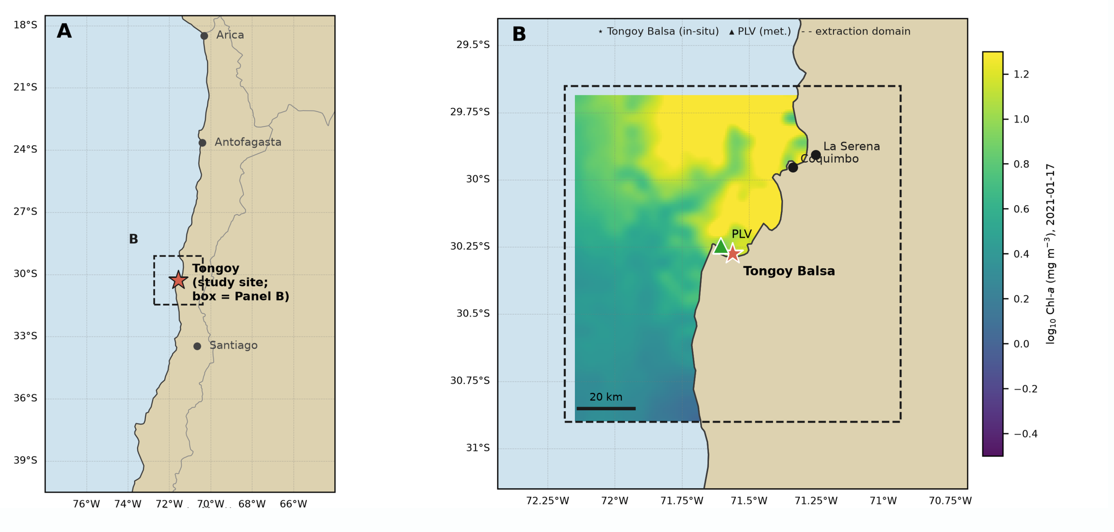
A leakage-controlled benchmark for reconstructing a patchy coastal biological record

BACKGROUND — coastal signal and target

One point buoy inside a patchy coastal field

Tongoy Balsa is one point sensor inside a patchy coastal upwelling field: it observes daily in-situ Chl-*a* in a system where blooms can be short-lived and spatially heterogeneous. Regional predictors (satellite products, meteorology from a nearby station, reanalysis) can be useful context to explain the upwelling mechanism and thus help predict the Chl-*a* value. The challenge is local biological reconstruction, performing a time series imputation with the help of covariates explaining the physical mechanism.



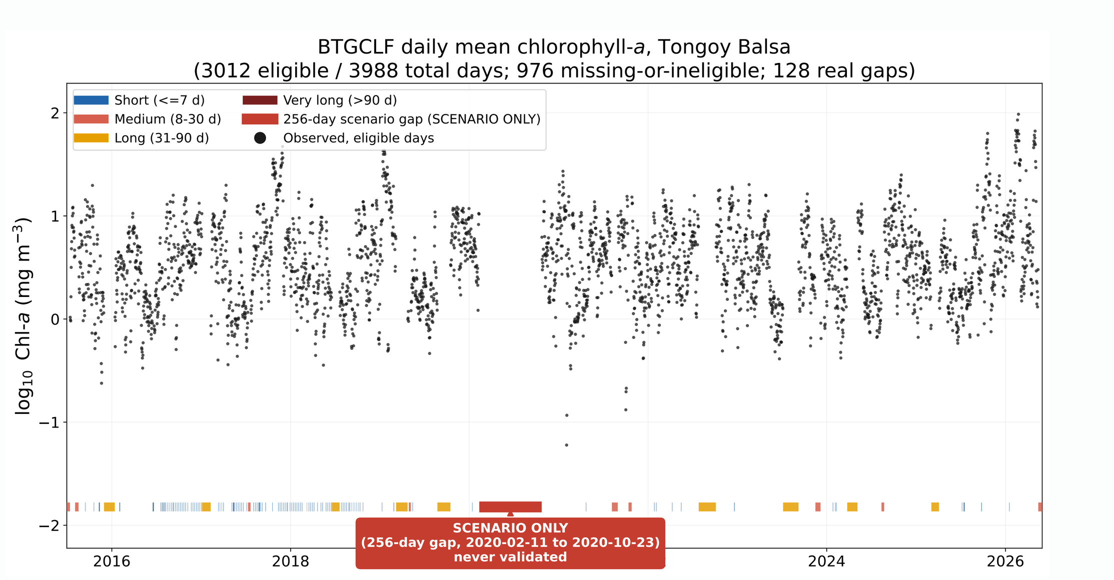
Site, PLV meteorological forcing station, extraction domain, and real satellite Chl-*a* context.

The Punta Lengua de Vaca meteorological station lies inside an active upwelling cell of the Humboldt system: wind-driven offshore transport brings cold, nutrient-rich water to the surface and can trigger short and intense blooms. Coarse-resolution satellite and reanalysis products average over scales of kilometres to tens of kilometres, so they can miss the bay-scale timing of a bloom that the buoy resolves at daily, point-source resolution. That mismatch in spatial scale is exactly why external products are admitted as covariates but carefully analysed in the model deployment to understand whether they improve or not the performance.

Target record and validation envelope

3988	3012	976	128
daily records 2015–2026	eligible days ≥18 valid h	missing / ineligible days	real gaps max 256 d

Item	Definition
target	daily mean in-situ Chl-*a*
eligibility	≥18 valid hourly observations
transform	$\log_{10}$ Chl-*a*
score	log-space MAE
validation	artificial gaps, L=1–60 d
outside support	256-d real gap, scenario-only



Information channels and evidence status

The considered models required different kind of inputs, thus the feature engineering process is specific to a model choice. Linear models and tabular trees require explicit lags, rolling summaries, anomalies and availability flags in order to capture at best the possible non linearities behind the upwelling mechanism; TS-ICL is already designed for time series and thus consumes raw aligned sequences. It was indeed proven that redundant physical stacks can dilute TS-ICL conditioning rather than help it.

Channel	Role	Evidence status
target history	local biological anchor	strongest local context
satellite Chl-*a*	biological proxy	strongest auxiliary (TS-ICL)
wind / upwelling	forcing pathway	genuine isolated signal
SST / cloud / PLV	thermal / meteo context	no placebo-survival effect
currents / transport	retention/flushing hypothesis	negative, tested products
calendar	seasonal baseline	baseline context only

Each channel above was tested by a placebo dissection to understand whether the covariate was truly helpful or not: the same predictor family is supplied either correctly aligned in time or deliberately scrambled (wrong lag, season-shuffled, year-shifted, permuted), and a family only counts as a genuine signal if it beats every one of those four placebo controls. This is what separates "satellite proxy helps" from a coincidental correlation that a less careful test would have accepted.

Stacking covariates is not free: with TS-ICL, the best imputing model in our study, a curated low-redundancy physical set (7 columns) scores a mean MAE of 0.1988, essentially tied with the target-only baseline (0.2014), while the full redundant physical union (79 columns) scores 0.2237 – clearly worse.

This means that more covariates channels without a redundancy check not only affect TS-ICL's inference time, as one could expect, but make conditioning noisier too.

Claim boundary: real-gap discipline

The same four-way distinction recurs across every result on this poster and is restated here as a single reference table.

artificial gaps	hidden truth, validation-grade
real gaps	no truth, plausibility/output only
256-day gap	outside validation, scenario-only
event labels	evaluation-only, cannot route real-gap methods

Whenever a figure on this poster shows a real-gap reconstruction, treat it as an illustration of model behaviour, not as evidence for or against any method.

It would be easy to point at a plausible-looking real-gap curve and call it evidence. The validation protocol on this research exists specifically to prevent that move. Moreover, validation on long gaps is harder: having many short gaps in the original time series, the longer the desired artificial gap, the harder to find candidate windows that don't include missing days.

METHODS — validation and model ladder

Hide, reconstruct, score

All method claims come from artificial gaps: observed Chl-*a* values are hidden, reconstructed with admissible context, then scored against withheld truth. Thus artificial gaps rank methods, while real gaps, the actual target and objective of this project, only show plausible behaviour.

observed daily series → hide artificial gap → reconstruct allowed context → score withheld truth

LOCO (Leave-One-Cluster-Out) cross-validation removes any training row whose target-history window overlaps the hidden gap; this is the safeguard against target leakage.

Uncertainty is computed by a gap-cluster bootstrap rather than by resampling individual days: whole gaps are resampled together, so days drawn from the same withheld interval cannot be treated as independent evidence.

The 681-gap canonical pool spans nine validated lengths (L=1, 3, 7, 10, 14, 21, 30, 45, 60 days), stratified by season and by event/non-event status, so a single method cannot look strong purely by being tested on an easy mix of gap lengths or calm-water periods.

Model ladder: how each model reconstructs a gap

Family	What it sees	Mechanism	Limitation
Linear interpolation	two gap-edge observations	straight line in $\log_{10}$ space	misses peaks inside the gap
GP / Kalman	local target neighbourhood	temporal / state-space smoothing	GP short-gap only; Kalman near interpolation
External tabular	engineered satellite/SST/wind/-calendar rows	tree ensemble predicts each hidden day	did not beat interpolation
Gap-edge residual	interpolation + pre/post summaries	learned correction over interpolation	retrospective; event-weak
TS-ICL satellite-proxy	**target sequence + satellite proxy**	**zero-shot probabilistic imputation**	**leading candidate but flattens peaks**

Why TS-ICL is the key methodological step. The first four methods either connect the gap edges, smooth the local target series, or convert each hidden day into hand-engineered predictors. TS-ICL is different: it keeps reconstruction as a **masked sequence problem**. The visible buoy Chl-*a*, the missing interval, calendar position and covariates are read together as one temporal context.

ts-icl

visible target buoy Chl-*a* | masked gap | aligned covariates satellite / forcing | calendar time position

⇓ pretrained probabilistic sequence model

median imputation + quantile bands / uncertainty

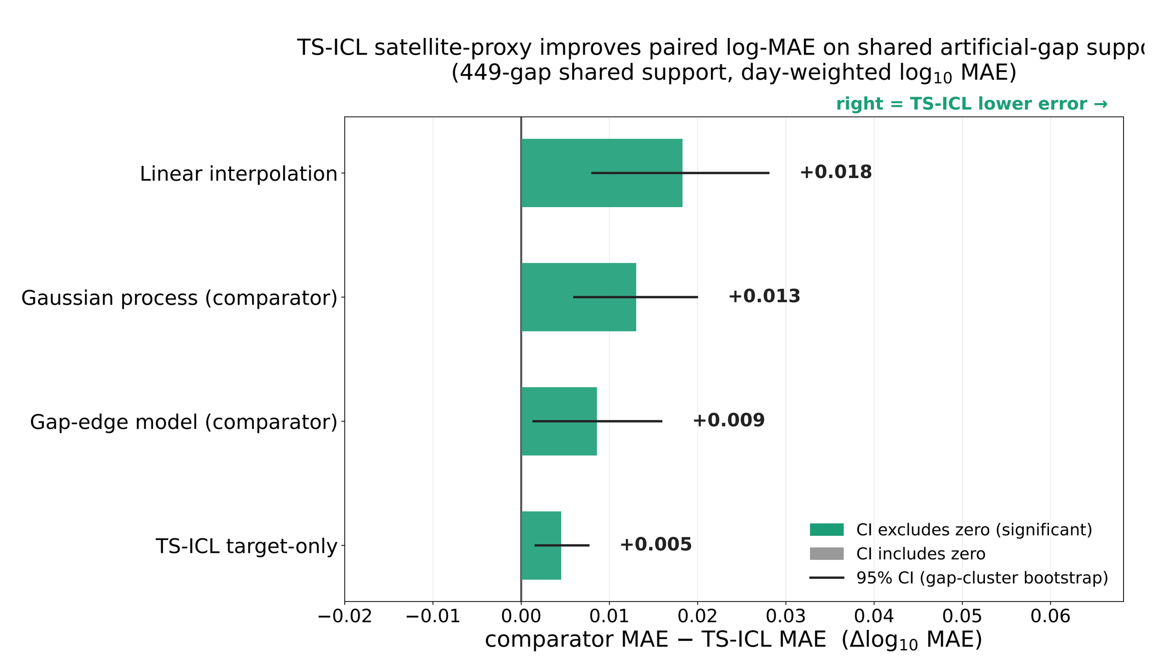
ICL = in-context learning. The model is not retrained on Tongoy. It adapts at inference time by reading the visible part of the current sequence: boundary behaviour, local variability, seasonality, and aligned covariates around the gap.

Zero-shot use. The pretrained model is applied directly to this buoy record, without fitting new local weights. The adaptation happens through the input context, not through a Tongoy-specific training stage.

Probabilistic output. TS-ICL does not only return one deterministic curve. It can provide a distribution of plausible trajectories, summarized by a median prediction and quantile bands. This is important for climate and coastal applications: a reconstruction should communicate both the expected Chl-*a* value and the uncertainty around it, especially when a gap may contain an unobserved bloom.

Why it matters here. TS-ICL is useful both as an imputer and as a covariate-audit tool. If real satellite Chl-*a* improves masked-sequence reconstruction while wrong-lag, shuffled, or permuted versions do not, the gain is evidence for time-aligned biological information.

Central benchmark result



Bars show comparator MAE − TS-ICL MAE on the shared artificial-gap support: right / positive means TS-ICL satellite-proxy has the lower error.

Negative results that matter

A benchmark is only as credible as the failures it is willing to report.

- external tabular features did not beat interpolation as a direct imputer;
- gap-edge residual helps mainly longer / smoother gaps, and is non-monotonic: L=10 d actually performs worse than plain interpolation, recovering only at $L \geq$45 d;
- GP is useful at L=1–3 but offers no external covariates, so its failure mode is purely temporal;
- currents/transport, expected to help reconstruction, are negative under two architectures. The coastal scale of the problem is not helped by predictors on coarse grids;
- redundant physical stacks hurt TS-ICL;
- event gaps remain harder than non-event gaps across the entire ladder.

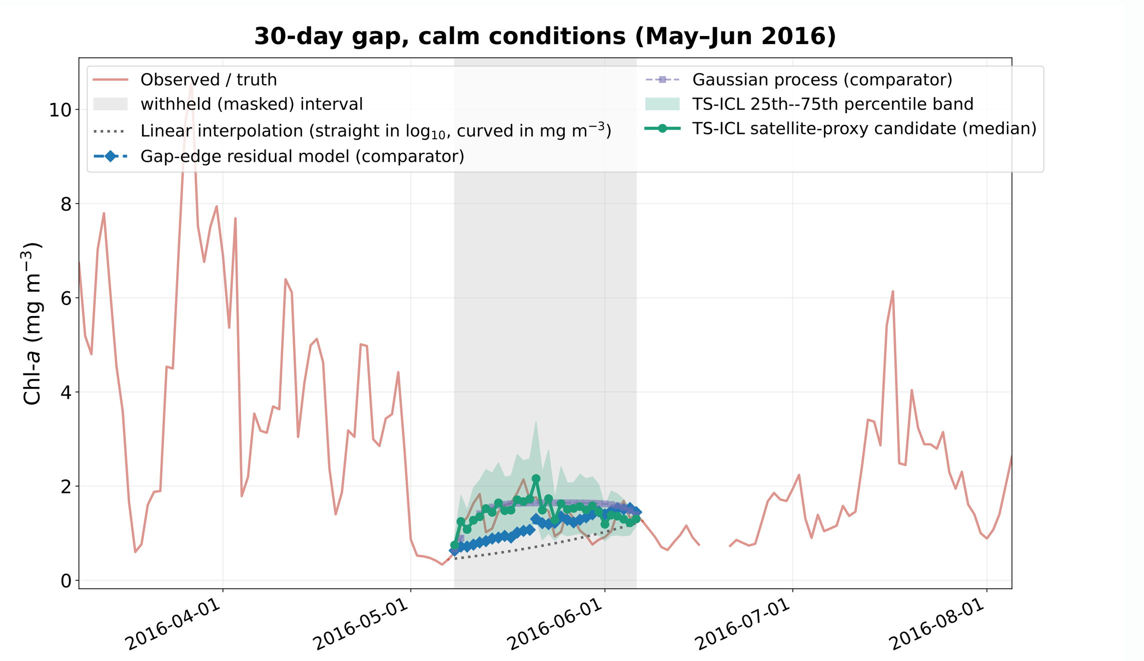
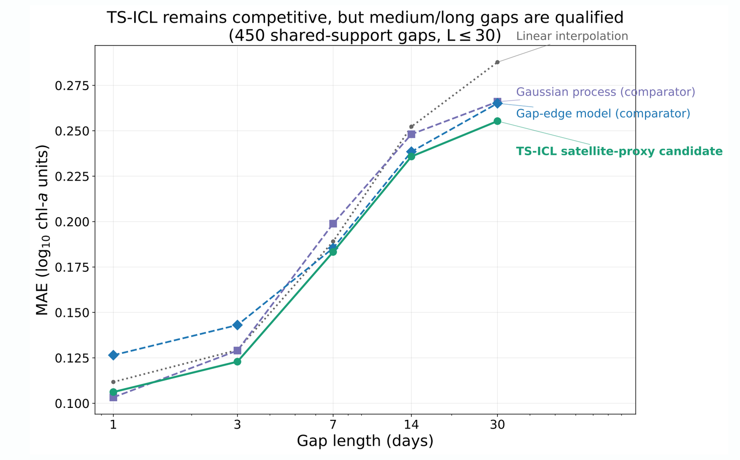


Illustration of the "helps mainly longer/smoother gaps" result: a 30-day artificial gap under calm conditions (May–Jun 2016, true max 2.14 mg m^{-3}). Gap-edge residual and TS-ICL satellite-proxy both stay close to truth across the full 30 days, while linear interpolation drifts more visibly.

RESULTS — mechanisms, events, real gaps

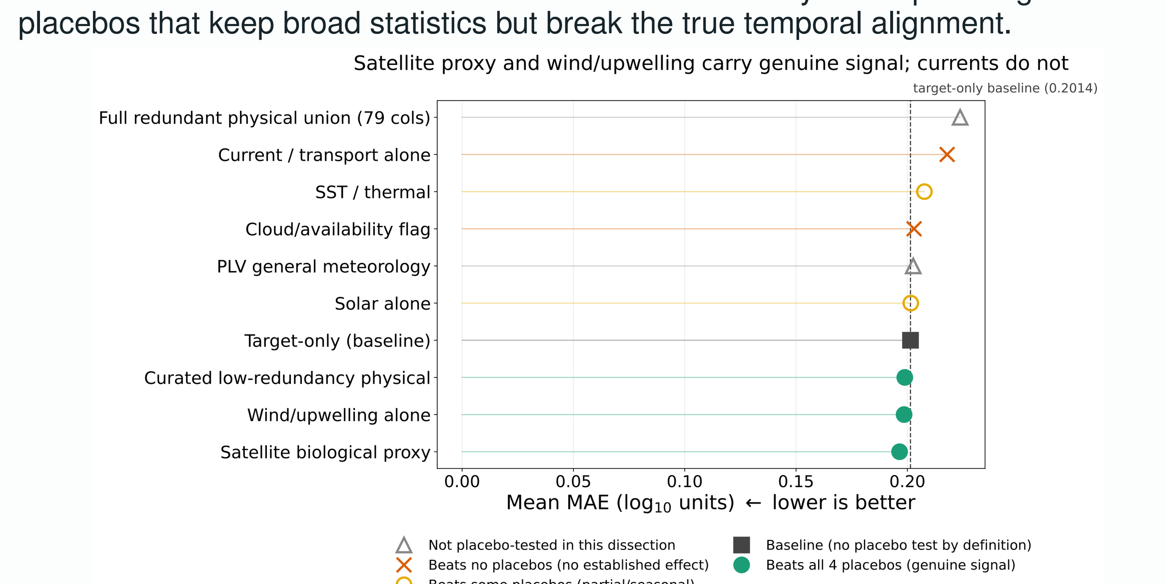
Gap-length behaviour by validation length

Comparator rankings vary with gap length: a pooled advantage is not universal superiority at every length. TS-ICL satellite-proxy has the lowest log-MAE from L=1 to 30 days, and its margin over interpolation, GP, and the gap-edge model widens as the gap lengthens.



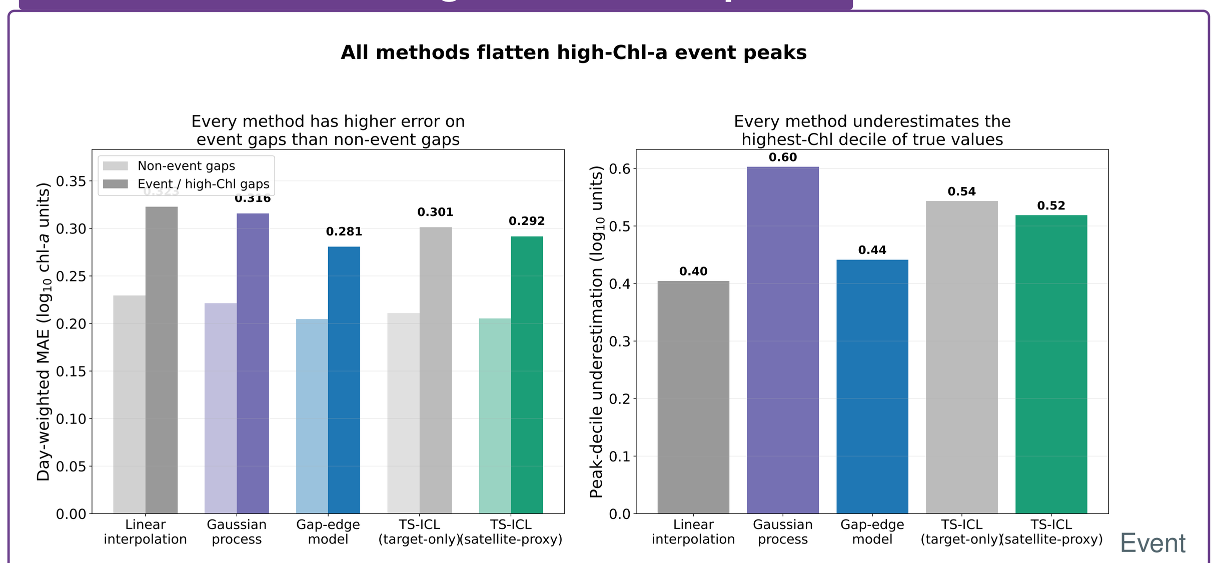
Covariate mechanism evidence

TS-ICL is also a covariate-audit tool: each candidate family is compared against placebos that keep broad statistics but break the true temporal alignment.



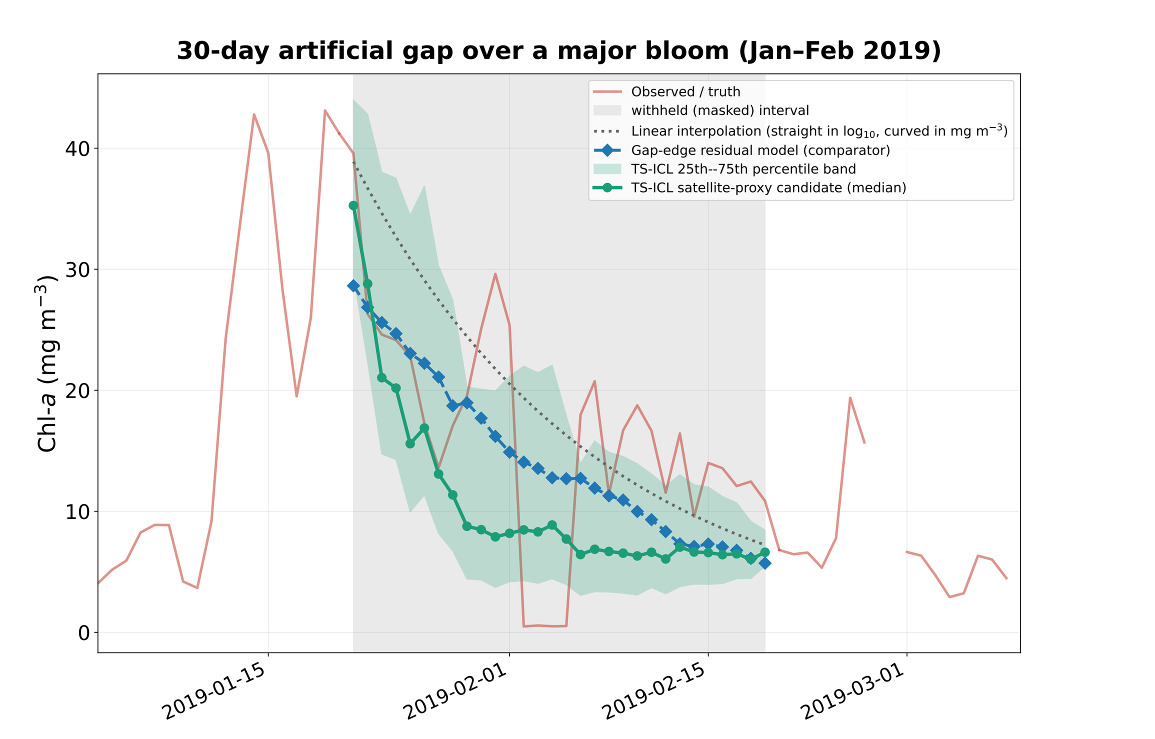
Family	TS-ICL result	Verdict
satellite proxy	best auxiliary	genuine signal
wind/upwelling	smaller, survives placebos	genuine, isolated
currents/transport	worse, no placebo support	negative

All methods flatten high-Chl-*a* event peaks



Event gaps have higher MAE than non-event gaps; the highest-Chl decile is underestimated by every method. TS-ICL is best among tested families on event gaps but does not solve peak recovery – high-Chl event reconstruction is the unresolved regime.

Same hidden truth, different reconstruction assumptions

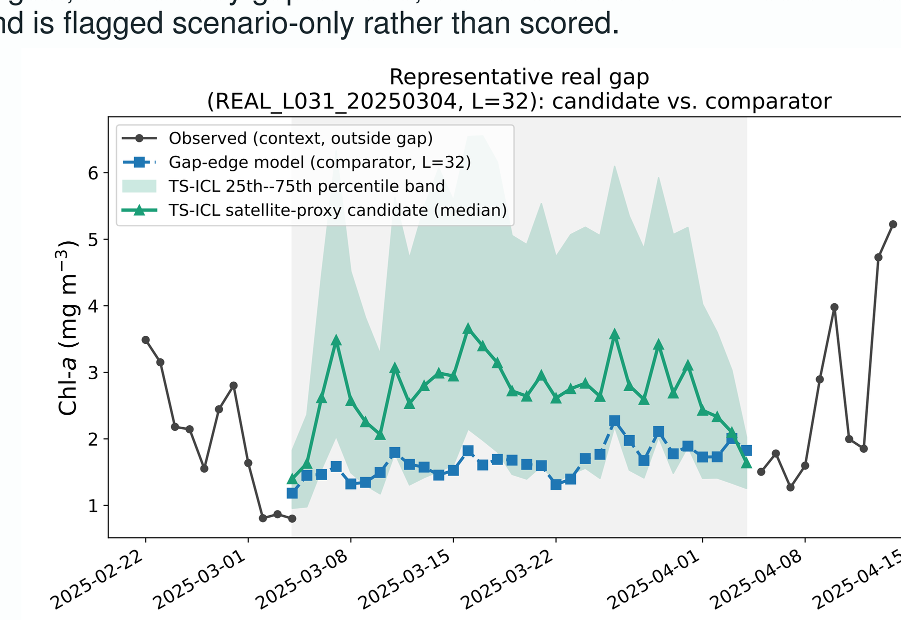


Method	Sees	Mechanism	Hidden-bloom behaviour
Interpolation	boundary values only	straight line, $\log_{10}$	misses internal peaks
Gap-edge residual	interp. + pre/post summaries	learned correction	boundary-conditioned, event-weak
GP smoother	local neighbours	smooths local structure	suppresses sharp peaks
TS-ICL satellite-proxy	target hist. + satellite proxy	zero-shot sequence imputation	follows state, compresses amplitude

Event lesson: the hidden peak is underdetermined when it rises and decays entirely inside the masked interval – boundary-conditioned and sequence-native methods alike compress it.

Real-gap output: plausibility only

Real gaps have no withheld truth; output is plausibility only and cannot rank methods. The longest, the 256-day gap of 2020, sits far outside the L=1–60 d validated envelope and is flagged scenario-only rather than scored.



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